Detection-Theoretic Decision Rules for AI-Agent Monitors

Overall summary of three rounds of experiments, a white-box pilot, and two validity audits

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In one paragraph. AI-control monitoring has improved its *monitors* (LLM judges, prompts, probes) much more than its *decision layer*: the rule that turns per-step scores into audit decisions. Almost every system takes the maximum over steps and compares it with one global honest-run quantile. Starting from detection theory (Neyman–Pearson, CFAR, CUSUM, sequential tests), we asked three questions:

1. Do accumulating rules beat max at a matched false-alarm budget?
2. Should thresholds be set per context?
3. Can an attacker who knows the detector game it?

We also ran a white-box pilot that feeds probe scores from an open model into the same decision layer. Over three rounds of experiments, followed by two independent audits that re-derived the headline numbers and corrected several claims, the surviving picture is:

- **max** (or top-3 sum) remains the robust default;
- accumulation helps only in narrow settings;
- **CFAR is the wrong per-context rule against an attacker who chooses where to act**, while damage-equalizing (Stackelberg) allocation is better at a 5% budget;
- **LLM rewriting of attacks dominates detector-aware red-teaming**, so an uninformed-rewrite control is essential;
- a **probe read on a follow-up question by an open trusted reader beats a cheap LLM monitor at 1% FPR**, and probe triage nearly doubles detection while calling the LLM on 13% of steps.

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1 What was done

1. **Literature review.** We read 92 arXiv papers in full and 101 lab and community sources (Redwood, Apollo, UK AISI, Anthropic, OpenAI, GDM, METR, security vendors), and wrote a separate review of white-box monitoring. We ranked 10 ideas and ran the top three plus a white-box pilot.
2. **Infrastructure.** The private repository `gregkocher/agent-monitor-detection` contains:
   - loaders for MonitoringBench, Honest-550, the LinuxArena 12-monitor sweep, aksh-n per-step labels, ScaleAI MRT, persistent-state VibeCoding and MOLE;
   - a per-step OpenRouter scoring harness (logprob-expected scores, resumable cache);
   - a detector library calibrated at matched false-alarm budgets;
   - a spend ledger;
   - a trusted-reader white-box pipeline on RunPod GPUs.
3. **Three rounds of experiments** per idea; Section 3 lists them.
4. **Two independent validity audits.** Each re-derived the headline numbers with its own minimal code and checked calibration leakage, grouping, confounds, onset labels, attack validity, spend matching, selection effects and report/table consistency. A fixer re-analyzed every affected result with corrected methods:
   - a single-resampling paired bootstrap clustered by task or account, including calibration noise;
   - true spend matching;
   - like-for-like attacker arms;
   - strict validity;
   - prompts matched except for the variable under test.

   Corrections are tracked claim by claim in `results/fixes/corrections.csv` and `results/audit2/corrections.csv`.

2 Results at a glance

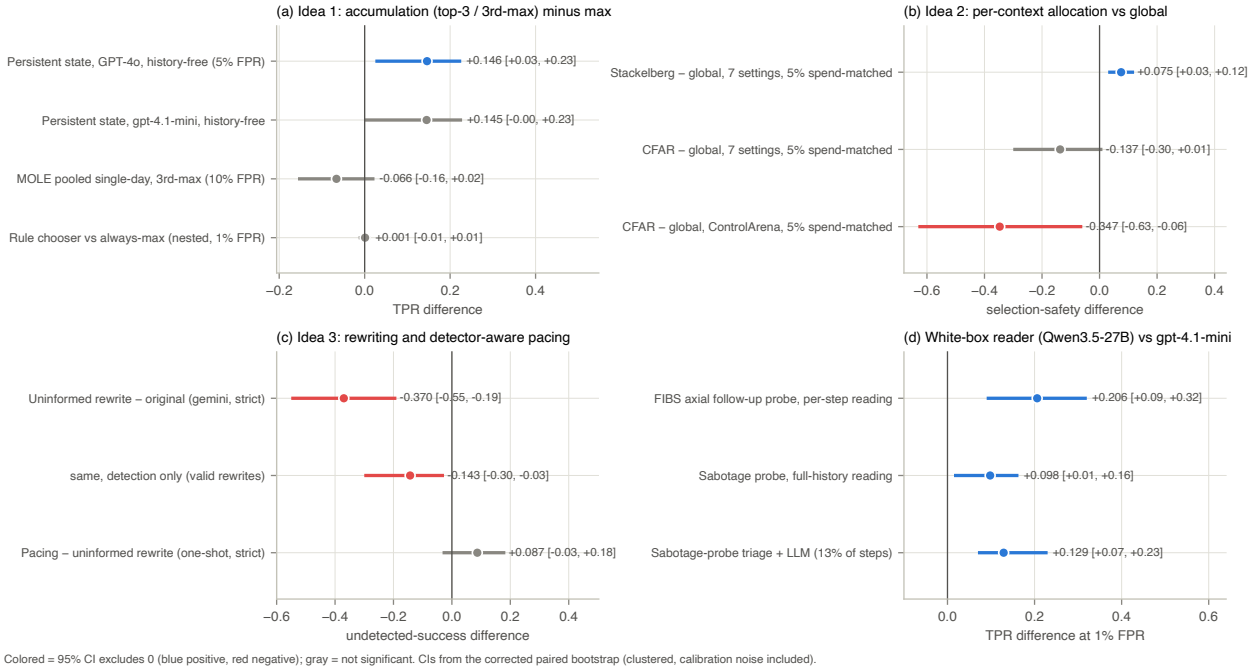


Figure 1: Audit-verified effects with corrected 95% CIs. Colored = CI excludes zero; gray = not significant. Sources: `results/fixes/corrections.csv`, `results/audit2/corrections.csv`, `results/t2/tables.md`, `results/r3_t3/paired_contrasts.md`, `results/wb/tables.md`.

3 The four research directions

3.1 Idea 1: accumulation vs max at a matched budget

Ran.

- **Round 1:** 46 real score-stream sets (aksh-n, LinuxArena sweep, MRT, persistent state), plus a controlled experiment: MonitoringBench re-scored per step by `gpt-4.1-mini` with full history vs none.
- **Round 2:** cross-session re-scoring of persistent state with three new monitor families; MOLE (30-day accounts); a data-driven rule chooser.
- **Round 3:** all 8 MOLE corpora under one common monitor, with activity and account-centering controls; long unreset streams; four "re-flagging" monitor designs.

Holds.

- Max, 2nd-max and top-3 sum are strong defaults. Textbook sequential tests (CUSUM k=0.5, Shiryaev–Roberts, e-processes) lose to them on almost every set, partly because most honest steps score exactly 0.
- Cross-session accumulation with a history-free per-PR GPT-4o monitor: +0.146 [0.025, 0.226].
- On long unreset streams a forgetting rule (CUSUM) beats max: 0.71 vs 0.58 at $\text{ARL}_0 = 100$ PRs.

- On MOLE, no accumulation rule beats max on any of the 8 corpora (pooled 3rd-max −0.066 [−0.156, 0.023]).

Weakened.

- The `gpt-4.1-mini` cross-session gain: +0.145 [−0.001, 0.228].
- The repeat-spike mechanism: descriptive only once dependence between cells is accounted for.

Dropped.

- The rule chooser's gain (+0.001 with nested calibration).
- "Accumulators gain more with history". It was a prompt-instruction confound, and vanishes with prompts matched except for history.
- The Gemini and Qwen cross-session gains.
- The round-2 MOLE gain. It was never significant, and was largely an account-activity effect.

3.2 Idea 2: per-context thresholds

Ran.

- **Round 1:** heterogeneity measurement; budget-matched allocation (global, CFAR, Mondrian, partial pooling, max-detection, Stackelberg); context-selecting and poisoning attackers.
- **Round 2:** transfer across attack policies and datasets; multi-context poisoning; severity-weighted damage; probing attackers; white-box and strong-monitor panels (E5).
- **Round 3:** Watcher-like severity tiers.

Holds.

- Honest-score heterogeneity is large across deployments and small within one. Probe scores are far more context-dependent than LLM scores (τ up to 3.79 vs ≤ 0.46).
- CFAR is the worst rule against an attacker who chooses its context, and a few alarm-feedback probes break it.
- Stackelberg beats global at 5% (0.97 vs 0.89).
- Partial pooling handles small per-context samples.
- Watcher-like severity tiers lose 0.06–0.12 to a plain suspicion threshold at 5% and cannot run at 1%.

Weakened or dropped.

- The 1% Stackelberg gain is not established, and the ControlArena 1% gain is dropped.
- "CFAR doubles probe TPR" was a spend artifact: at equal spend the gain is 1.3–1.5×.
- Stackelberg fails when fit on different side tasks.
- The leave-one-context-out poisoning defense breaks when poison is spread across contexts. A frozen trusted reference holds, but over-spends under genuine drift.

3.3 Idea 3: detector-aware attackers

Ran.

- **Round 1:** a `gpt-4.1` rewriter told the monitor prompt and detector, with `gpt-4.1-mini` as monitor.

- **Round 2:** a strong `gpt-4.1` monitor; an anti-condensation pacing attacker; an uninformed-rewrite control.
- **Round 3:** a different-family rewriter (Gemini 2.5 Pro); a check that fabricated outputs don't leak cues; realized FPR; replay feasibility.

Holds.

- Any LLM rewrite makes attacks easier to catch, whether the rewriter is the monitor's own model family or a different one: -0.37 [-0.55, -0.19] for the Gemini rewriter. About 60% of that drop is broken attacks; detection only is -0.14 [-0.30, -0.03].
- Detector-aware pacing shows no significant strategy effect when compared one-shot against one-shot ($+0.087$ [-0.032, 0.183]).
- Max, CUSUM k=1.5 and top-3 sum hold nominal FPR. The multi-chart OR over-alarms (0.017–0.020 at a nominal 0.01).
- The covert-adversary $\sqrt{\text{ARL}}$ law is rejected in this setting: detection hinges on single conspicuous steps.

Retracted.

- Round 1's "aware attacks backfire" (there was no control).
- Round 2's "+0.21 pacing gain" (an adaptive arm was compared with a one-shot control).

3.4 White-box pilot

Ran.

- A Qwen3.5-27B trusted reader on two H200 pods, reading each transcript per step or with full history.
- Signals:
  - a synthetic sabotage probe;
  - Apollo's deception probe;
  - FAR.AI FIBS probes, read on the step tokens and on a follow-up question;
  - a J-lens watch-list.
- All signals passed through the decision layer (decision rules, per-context thresholds, LLM fusion, detector-aware rewrites).
- Data: MonitoringBench, LinuxArena, and MOLE policy-vs-reader.

Holds.

- The FIBS follow-up probe, read per step, beats `gpt-4.1-mini` at 1% FPR: +0.21 [0.09, 0.32]. It is the only contrast that survives Holm correction across 18.
- Sabotage-probe triage plus LLM (13% of steps) reaches 0.30 vs 0.17 at matched realized FPR.
- Pre-onset attack steps score like honest steps, so the probe is not leaking the attack in advance.

Weakened.

- The full-history sabotage probe gain (+0.10): Holm p 0.22, and present only on one attack agent model.
- The full-history follow-up probe and the fusions built on it: its threshold does not transfer across tasks (2.05% realized FPR at a nominal 1%).

- The LinuxArena ranking and the round-1 rewrite comparison: their score files were lost in the final pod sync, so they are now unverifiable.

Not supported.

- J-lens watch-list readouts (AUROC ≤ 0.72).
- Reading with the agent's own model beating the trusted reader on the sabotage probe.

4 Recommendations for practitioners

1. **Default trajectory rule:** max with a length-aware null where lengths differ; top-3 sum or CUSUM with $k \approx 1.5$ as alternatives. Calibrate any OR combination *jointly*, and verify realized FPR on held-out honest runs.
2. **Accumulate across sessions only after checking** that repeated high scores are informative for your monitor and data. Prefer history-free per-session judgments, and use forgetting rules on long streams.
3. **Per-context thresholds:**
   - measure heterogeneity first;
   - never use CFAR when attackers can choose contexts;
   - use Stackelberg (damage-equalizing) allocation fit on attack data that resembles the threat, with shrinkage;
   - use partial pooling for small samples;
   - use a trusted frozen reference against poisoning.
4. **Severity tiers** (Watcher-style) should be validated against an honest hit-rate before being wired in. Rank by suspicion, with one budget-matched threshold per deployment.
5. **Red-teaming with LLM rewrites** must include an uninformed-rewrite control and strict validity checks; otherwise the rewrite artifact masquerades as a strategy effect.
6. **White-box:** a probe read on a follow-up question is a cheap, strong complement to LLM monitors, and an efficient triage stage. Probe thresholds are strongly context-dependent, so calibrate per context.

5 What is new (to our knowledge, per the literature review)

- A matched-budget comparison of sequential and anytime-valid statistics vs max on real AI-control score streams. It answers MonitoringBench's stated open question, largely in favor of max.
- Evidence, with corrected CIs, that cross-session accumulation of history-free judgments can beat max for some monitors. The source paper tested only a failing k-strikes rule.
- A per-context threshold study for control monitors: CFAR's failure under attack selection, the Stackelberg alternative, poisoning-robust baselines, and Watcher-like severity tiers.
- Evidence that LLM rewrite artifacts dominate detector-aware red-teaming, and that detector-aware pacing shows no measurable strategy effect once controlled.
- A trusted-reader white-box pipeline evaluated through the same decision layer, showing a follow-up-question probe beating a cheap LLM monitor at 1% FPR, and efficient probe triage.

6 Validity process and errata

The first audit found 2 critical, 9 major and 13 minor issues; the second found 4 major and 10 minor. All point estimates reproduced with independent code. The main corrections relative to the October 21 summary:

Original claim (Oct 21)	Corrected	Verdict	Source
Cross-session gain +0.15 [0.12, 0.17]	GPT-4o +0.146 [0.025, 0.226]; gpt-4.1-mini [−0.001, 0.228]	holds / weakened	fixes C2
Rule chooser +0.023	+0.001 [−0.014, 0.013] (nested)	dropped	fixes M6
Accumulators gain more with history	prompt confound; DiD ≈ 0 with matched prompts	dropped	fixes M4
CFAR doubles probe TPR at 1%	1% not estimable; 1.3–1.5× at equal spend	dropped / weakened	fixes C1
Aware attacks backfire (round 1)	rewrite artifact (control arm)	dropped	fixes M3
Pacing vs control +0.21	one-shot vs one-shot +0.087 [−0.032, 0.183]	weakened (n.s.)	fixes M1
Gemini rewrite cost −0.37	detection only −0.14 [−0.30, −0.03]	weakened	audit2 A4
MOLE accumulation gain +0.13	never significant; no rule beats max on 8 corpora	dropped	r3_t1, audit2
History-aware follow-up probe lead	threshold over-alarms (2.05% at 1%)	weakened	audit2 A1

7 Budget, compute and artifacts

Item	Spend	Notes
Round 1 (OpenRouter)	$195.56	Ideas 1–3 initial experiments
Round 2 (OpenRouter)	$455.61	Idea 1 $147.89, Idea 3 $304.77, white-box $2.95
Round 3 (OpenRouter)	$294.20	Idea 1 $161.36, Idea 2 $21.82, Idea 3 $111.02
Audit fixes (OpenRouter)	$25.31	matched-prompt re-scoring
OpenRouter total	**$970.69**	MATS account; no personal-key spend
RunPod GPU	≈$72	H200 pods for the white-box reader; all deleted

- **Repository:** https://github.com/gregkocher/agent-monitor-detection (private). Code in `amd/` and `tracks/`, results in `results/`, write-ups in `docs/`.
- **Run history:** `EXPERIMENT_LOG.md` has the full dated history.
- **Detailed reports:** `docs/reports/01_Idea1_Matched_Budget.pdf`, `02_Idea2_Per_Context_Thresholds.pdf`, `03_Idea3_Aware_Attacker.pdf`, `04_WhiteBox_Pilot.pdf`.
- **Literature:** `docs/research/` has the literature summary, idea ranking and reading cards.

8 Limitations and next steps

Limitations.

- **Weak monitors and small samples.** The re-scored monitors are mostly cheap (`gpt-4.1-mini`/`gpt-4.1`). Many attack sets are small (35–150 base attacks), and honest sets are 200–550 runs. Low-FPR estimates are noisy.
- **Attacks were not executed.** Rewritten attacks were validated by LLM judge plus target-literal checks only.

- **Lost white-box files.** Some white-box score files were lost, and the LinuxArena reader results cannot be re-derived.

Recommended next steps, in priority order:

1. **Execution replay of attacks** ($0 API, laptop Docker). The infrastructure works. It needs:
   - structured tool calls re-extracted from the source logs;
   - a rewriter that outputs structured calls.
2. **White-box follow-ups:**
   - a preregistered, Holm-controlled replication of the follow-up probe;
   - regenerating the lost score files;
   - a policy-activation arm with account-matched benign sessions.
3. **Cross-session accumulation** with more monitors and longer real multi-session data, using the corrected bootstrap throughout.
4. **Stackelberg allocation at 1%**, with more honest runs per context so 1% is reachable.